

For reading on complex numbers, please refer to Appendix B (Pages 689-700) of [Structure and Interpretation of Signals and Systems](#). In this note, we will show solutions to some of the example problems from this book to help you test your knowledge.

0.1 Complex Number Problem Examples

0.1.1 Example 1

Depict the following complex numbers graphically:

$$i, -2, -3 - i, -1 - i \quad (1)$$

Solution:

To plot complex numbers represented in Cartesian form, we plot the real part on the x-axis and the imaginary part on the y-axis.

Here are the real and imaginary parts of each of these complex numbers:

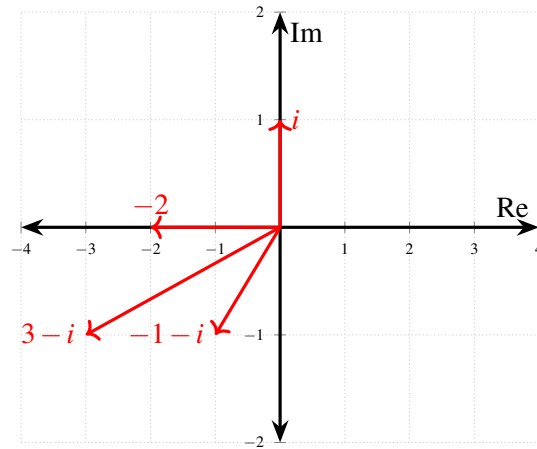
$$\operatorname{Re}\{i\} = 0, \operatorname{Im}\{i\} = 1 \quad (2)$$

$$\operatorname{Re}\{-2\} = -2, \operatorname{Im}\{-2\} = 0 \quad (3)$$

$$\operatorname{Re}\{-3 - i\} = -3, \operatorname{Im}\{-3 - i\} = -1 \quad (4)$$

$$\operatorname{Re}\{-1 - i\} = -1, \operatorname{Im}\{-1 - i\} = -1 \quad (5)$$

Thus, we plot the complex numbers in the plot below:



0.1.2 Example 2

Simplify the following expression:

$$e^{2+\pi i} \quad (6)$$

Hint: use Euler's formula, which states that $e^{i\theta} = \cos(\theta) + i\sin(\theta)$.

Solution:

For this problem, let's first split the exponent (recall that Euler's formula works for $e^{i\theta}$ and thus cannot be applied directly to this problem due to the exponent having a real part):

$$e^{2+\pi i} = e^2 e^{\pi i} \quad (7)$$

The e^2 part is just some value ($e^2 \approx 7.39$ to be specific) and thus does not need to be simplified.

To simplify $e^{\pi i}$, we can use Euler's formula:

$$e^{\pi i} = \cos(\pi) + i\sin(\pi) = -1 \quad (8)$$

This makes sense! If we drew a vector of length 1 in the direction of $\theta = \pi$ in the complex plane, we would reach the point -1 (being able to visualize this will help a lot with complex numbers)!

Thus, the simplified value is:

$$e^{2+\pi i} = e^2(-1) = -e^2 \quad (9)$$

0.1.3 Example 3

Express the following in polar coordinates:

$$2 - 2i, 2 + 2i, \frac{1}{2 - 2i}, \frac{1}{2 + 2i}, 2i, -2i \quad (10)$$

Solution:

Let's start with the first two. Recall that to find the polar form representation of a complex number z , you must find its magnitude $|z|$ and phase $\angle z$:

$$|2 - 2i| = \sqrt{2^2 + (-2)^2} = \sqrt{8} = 2\sqrt{2} \quad (11)$$

$$\angle 2 - 2i = \arctan\left(\frac{-2}{2}\right) = \arctan(-1) = -\frac{\pi}{4} \quad (12)$$

$$\rightarrow 2 - 2i = 2\sqrt{2}e^{-i\frac{\pi}{4}} \quad (13)$$

Remember to visualize the location of the complex number in the complex plane for the phase; since $2 - 2i$ is in Quadrant IV, the correct phase is $-\frac{\pi}{4}$, not $\frac{3\pi}{4}$.

Similarly for $2 + 2i$:

$$|2 + 2i| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2} \quad (14)$$

$$\angle 2 + 2i = \arctan\left(\frac{2}{2}\right) = \arctan(1) = \frac{\pi}{4} \quad (15)$$

$$\rightarrow 2 + 2i = 2\sqrt{2}e^{i\frac{\pi}{4}} \quad (16)$$

For the next two, notice that these are reciprocals of the first two complex numbers (1 divided by the complex number). Division is much easier to do in polar form and conveniently, we already calculated the polar form of these two complex numbers.

The calculation proceeds as follows:

$$\frac{1}{2 - 2i} = \frac{1}{2\sqrt{2}e^{-i\frac{\pi}{4}}} = \frac{1}{2\sqrt{2}}e^{i\frac{\pi}{4}} \quad (17)$$

$$\frac{1}{2 + 2i} = \frac{1}{2\sqrt{2}e^{i\frac{\pi}{4}}} = \frac{1}{2\sqrt{2}}e^{-i\frac{\pi}{4}} \quad (18)$$

For the last two, the magnitude of both complex numbers is clearly 2. For the phase, it is easiest to visualize the complex number in the complex plane.

$2i$ can be represented by a vector that points up in the complex plane, which corresponds to a phase of $\frac{\pi}{2}$.

$-2i$ can be represented by a vector that points down in the complex plane, which corresponds to a phase of $-\frac{\pi}{2}$.

Thus, the polar form of these complex numbers is:

$$2i = 2e^{i\frac{\pi}{2}} \quad (19)$$

$$-2i = 2e^{-i\frac{\pi}{2}} \quad (20)$$

$$(21)$$

0.1.4 Example 4

Find all possible $\theta \in [0, 2\pi)$ such that

$$\operatorname{Re}\{(1+i)e^{i\theta}\} = -1 \quad (22)$$

Solution:

First, let's convert $1+i$ to polar form; this will make the multiplication between the two complex numbers easier to interpret:

$$|1+i| = \sqrt{1^2+1^2} = \sqrt{2} \quad (23)$$

$$\angle 1+i = \arctan\left(\frac{1}{1}\right) = \arctan(1) = \frac{\pi}{4} \quad (24)$$

$$1+i = \sqrt{2}e^{i\frac{\pi}{4}} \quad (25)$$

Thus,

$$(1+i)e^{i\theta} = \sqrt{2}e^{i\frac{\pi}{4}}e^{i\theta} = \sqrt{2}e^{i(\theta+\frac{\pi}{4})} \quad (26)$$

Now, we can use Euler's formula to find an expression for the real part and set this equal to -1 :

$$\operatorname{Re}\left\{\sqrt{2}e^{i(\theta+\frac{\pi}{4})}\right\} = \sqrt{2}\cos\left(\theta+\frac{\pi}{4}\right) = -1 \quad (27)$$

$$\cos\left(\theta+\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} \quad (28)$$

$$\theta+\frac{\pi}{4} = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (29)$$

$$\theta = \frac{\pi}{2}, \pi \quad (30)$$

0.1.5 Example 5

Express the six distinct roots of unity, i.e. the six solutions to

$$z^6 = 1 \quad (31)$$

in Cartesian and polar coordinates.

Solution:

For these types of problems, it is usually useful to start with polar form (which makes sense since exponentiation is repeated multiplication and multiplication is easier in polar form).

Notice that we can write $1 = e^{i2\pi n}$, with any integer n . Using this polar representation of 1, the sixth root becomes a division of the exponent by 6:

$$z = (e^{i2\pi n})^{\frac{1}{6}} = e^{\frac{i2\pi n}{6}} = e^{\frac{i\pi n}{3}} \quad (32)$$

If we evaluate this for $n = 0, 1, 2, 3, 4, 5$, we find 6 distinct solutions for z (other values will result in repeated phases):

$$z = e^{i0}, e^{i\frac{\pi}{3}}, e^{i\frac{2\pi}{3}}, e^{i\pi}, e^{i\frac{4\pi}{3}}, e^{i\frac{5\pi}{3}} \quad (33)$$

Converting to Cartesian coordinates using Euler's formula:

$$z = 1, \frac{1}{2} + i\frac{\sqrt{3}}{2}, -\frac{1}{2} + i\frac{\sqrt{3}}{2}, -1, -\frac{1}{2} - i\frac{\sqrt{3}}{2}, \frac{1}{2} - i\frac{\sqrt{3}}{2} \quad (34)$$